



Lab 07: VLAN Segmentation

2960-24TT Switch · PC0 · PC1 · PC2 — Three VLANs

Lab #07

Topic: VLANs & Switching

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Create and name three VLANs on a single Cisco 2960-24TT switch.
- Assign PC0, PC1 and PC2 to different VLANs via access ports Fa0/1–Fa0/3.
- Verify same-VLAN devices communicate at Layer 2 normally.
- Confirm cross-VLAN devices are isolated without a router.
- Relate to Network+ objective 2.3: Purpose and use of VLANs.

2. Network Topology

A single 2960-24TT switch (Switch0) with no router. PC0 on Fa0/1 (VLAN 10), PC1 on Fa0/2 (VLAN 20), PC2 on Fa0/3 (VLAN 30). Without a Layer 3 device, cross-VLAN pings will fail — demonstrating VLAN isolation.

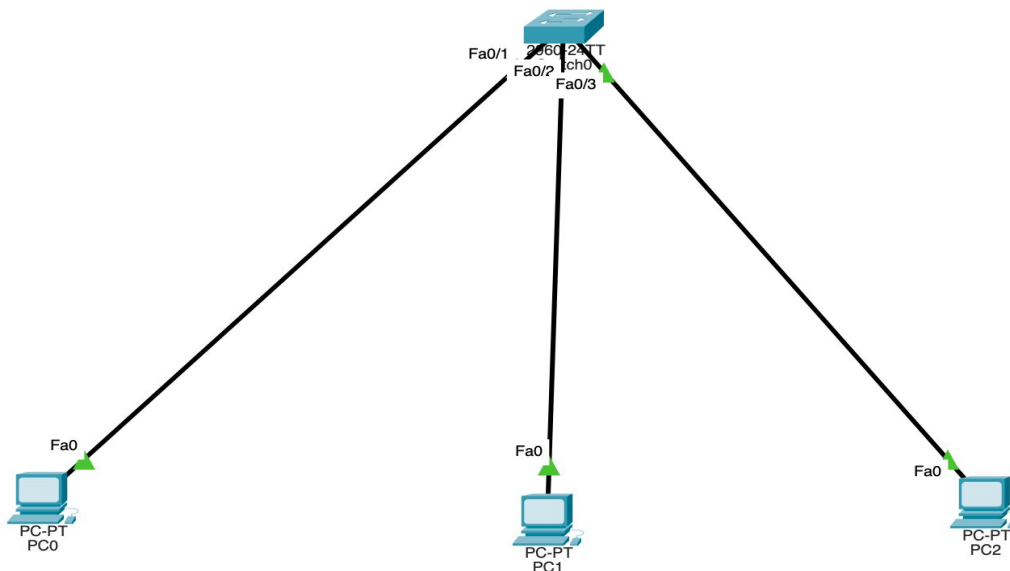


Figure 1 – Switch0 with PC0 (Fa0/1), PC1 (Fa0/2), PC2 (Fa0/3) on three VLANs

Device	Switch Port	VLAN ID	VLAN Name	IP Address	Mask
PC0	Fa0/1	10	SALES	192.168.10.10	255.255.255.0
PC1	Fa0/2	20	TECH	192.168.20.10	255.255.255.0
PC2	Fa0/3	30	MGMT	192.168.30.10	255.255.255.0



3. Configuration

3.1 Create VLANs

```
Switch>enable
Switch#configure terminal
Switch(config)#vlan 10
Switch(config-vlan)#name SALES
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name TECH
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name MGMT
Switch(config-vlan)#exit
```

3.2 Assign Access Ports

```
Switch(config)#interface FastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch#write memory
```

4. Verification

```
Switch#show vlan brief
PC0>ping 192.168.20.10 ! FAIL – different VLAN, no router
PC0>ping 192.168.30.10 ! FAIL – different VLAN, no router
```

5. Key Takeaways

- VLANs create logical broadcast domain separation on a single physical switch.
- Access ports carry traffic for ONE VLAN only — frames arrive and leave untagged.



- Without a Layer 3 device, VLANs are completely isolated — by design.
- VLANs improve security, reduce broadcast traffic and simplify management.
- **show vlan brief** is the fastest way to confirm port-to-VLAN assignments.